

Data Structure (C Programming)

25/08/2026

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Roll No : 15

Batch : B1

Prerequisite

Based on Array.

Q1. Write a C program to find the largest and smallest element in an array of integers.

Input:

Que 1] Based on Array
Write a C program to find the largest and smallest element in an array of integers.

```
#include <stdio.h>
int main()
{
    int a[100], size, i, max;
    printf("Enter Size of Array: ");
    scanf("%d", &size);
    for(i=0; i<size; i++)
    {
        scanf("%d", &a[i]);
    }
    max = a[0];
    for(i=0; i<size; i++)
```

```
{ if(a[i]>max)
{ max=a[i];
}}
printf("Maximum value: %d",
max min=a[0]
for(i=0; i<size; i++)
{ if(a[i]<min)
{ min=a[i];
}}
printf("Minimum element: %d", min);
return 0;
}
```

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Output:

```
PS C:\Users\Harshad Teli\Downloads\DS> gcc -o output1 Program.c
PS C:\Users\Harshad Teli\Downloads\DS> ./output1
Enter Size of Array:5
7
6
3
8
1
Minimum Value from the Array is 1
Maximum Value from the Array is 8
PS C:\Users\Harshad Teli\Downloads\DS> |
```

Q2. Write a C program to reverse the elements of an array.

Input:

Q.2] Write a C Program to reverse the elements of an Array

```
#include <stdio.h>
int main()
{
    int arr[10];
    printf("Enter Array Elements: \n");
    for(int i=0; i<10; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Reverse of the Array Element is \n");
    for(int i=9; i>=0; i--)
    {
        printf("%d\n", arr[i]);
    }
    return 0;
}
```

O/P: - Enter Array Elements with size of 5:

2 2
6
78
56
89

The reverse elements of the Array is:

89
56
78
6
2

Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ gcc -o output Classroom.c
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ ./output
Enter Array Elements with the Size of 5:
2
6
78
56
89
The reverse elements of the Array is :
89
56
78
6
2
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$
```

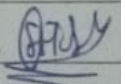

Q3. Write a C program to perform the addition of two matrices (2D arrays) of size 3×3 .

Input:

Q3] Write a C Program to perform the addition of two matrices (2D) of sizes 3×3 .

```
→ #include <stdio.h>
int main()
{
    int am1[3][3];
    int am2[3][3];
    int result[3][3];
    printf("Enter elements of first Array: ");
    for(int i=0; i<=2; i++)
    {
        for(int j=0; j<=2; j++)
        {
            scanf("%d", &am1[i][j]);
        }
    }
    printf("Enter elements of second Array: ");
    for(int i=0; i<=2; i++)
    {
        for(int j=0; j<=2; j++)
        {
            scanf("%d", &am2[i][j]);
        }
    }
    printf("The Addition of the Array is: \n");
    for(int i=0; i<2; i++)
    {
        for(int j=0; j<2; j++)
        {
            result[i][j] = am1[i][j] + am2[i][j];
            printf("%d\t", result[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

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Roll No: 15

Sign: 

Output:

Enter the elements of the Array1 with the Size 3X3:

Enter the element for the arr[0][0]=6

Enter the element for the arr[0][1]=23

Enter the element for the arr[0][2]=78

Enter the element for the arr[1][0]=96

Enter the element for the arr[1][1]=2

Enter the element for the arr[1][2]=5

Enter the element for the arr[2][0]=15

Enter the element for the arr[2][1]=36

Enter the element for the arr[2][2]=45

Enter the elements of the Array2 with the Size 3X3:

Enter the element for the arr[0][0]=10

Enter the element for the arr[0][1]=2

Enter the element for the arr[0][2]=36

Enter the element for the arr[1][0]=48

Enter the element for the arr[1][1]=2

Enter the element for the arr[1][2]=9

Enter the element for the arr[2][0]=3

Enter the element for the arr[2][1]=47

Enter the element for the arr[2][2]=2

The Addition of the Array is:

16	25	114
----	----	-----

144	4	14
-----	---	----

18	83	47
----	----	----